

Abhinav

✉ 1899abhinav@gmail.com | 🔔 schallten | 🌐 website

Education

Kalinga Institute Of Industrial Technology

Bhubaneswar, OD , IN

BTech in Computer Science and System Engineering

08/2024 - 06/2028

Projects

Mail Server

Go

08/2026

- Implemented SMTP (RFC 5321) and POP3 (RFC 1939) servers from the specifications, interoperable with standard clients like Thunderbird, with zero dependencies
- Persisted messages as RFC 822 .eml files using atomic writes (tmp → fsync → rename), preventing partial commits on crash
- Built per-user mailbox isolation with credential-backed authentication and graceful shutdown

Container Orchestration Platform

Go, Docker, Cloudflared

WebSockets, 04/2026 - 05/2026

- Engineered a multi-tenant container platform in Go provisioning isolated Linux environments in ~400ms via WebSocket terminal sessions with in-container Cloudflare tunnels
- Architected a lock-free concurrent scheduler launching 20 simultaneous containers in ~1.6s with goroutine-based cleanup and billing
- Hardened with defense-in-depth: cgroup resource limits, no-new-privileges, non-root sessions, and iptables metadata endpoint blocking

Distributed

Version

Control

Go, FastAPI, React

10/2025 - 12/2025

System

- Built a Version Control System CLI from scratch in Go, with SHA256 content-addressed storage tracking delta changes similar to Git's object database
- Added a React + FastAPI dashboard visualizing commit histories and branch topologies

Skills

Programming Languages: 🟢 Go, 🐍 Python, 🟢 C++, 🟡 JavaScript, 🟢 C, Odin

Tools and Frameworks: Docker, FastAPI, React, WebSockets, Linux (PTY, cgroups), Cloudflared, Git

Languages: English (Proficient), Hindi (Native)